SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

IEEE 830-1998 Standard | HCLTech BTech Curriculum (v1.0)

1. System Purpose & Functional Requirements

FR-1 (Multi-Sensor Acquisition): Samples DHT22, ACS712, MQ-135, and SW-420 every 3000ms.
FR-2 (ADC1 Channel Allocation): Analog sensors mapped exclusively to ADC1 (GPIO 32, 34, 35) to prevent Wi-Fi contention.
FR-3 (Noise Conditioning): 10-sample circular moving average filter attenuates electronic noise.
FR-4 (Autonomous Interlocks): Auxiliary fan energizes in < 15ms at Temp > 75°C; Conveyor halts at Vibe > 4.5 mm/s.
FR-5 (MQTT QoS 1 Telemetry): Publishes standardized JSON packets to HiveMQ broker with Last Will and Testament (LWT).
FR-6 (Cloud REST & WS API): Exposes authenticated REST endpoints and WebSocket stream with sub-100ms latency.
FR-7 (Predictive Maintenance): Unsupervised Isolation Forest flags early mechanical bearing degradation.

2. Non-Functional Requirements & Safety Constraints

NFR-1 (Performance): Edge safety shutoffs execute in < 15ms; API response time < 200ms.
NFR-2 (Reliability): 15-second hardware Watchdog Timer (WDT); Wi-Fi self-healing auto-reconnects in 3.0s.
NFR-3 (Security): HMAC-SHA256 JWT tokens with 15m expiration; Bcrypt password hashing; 0 hardcoded secrets.
NFR-4 (Cost Constraint): Total hardware unit cost constrained under ■2,000 INR (Delivered at ■1,840 INR).